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1 INTRODUCTION

In this report we present the steps that allowed us to design and implement a decision support system for a music platform. The project is based on two datasets: "Tracks" and "Artists". Starting from the data provided, several activities are carried out: a preliminary analysis of the datasets, the design of the Data Warehouse schema, the creation and population of the Data Warehouse, the definition and execution of analytical queries using SQL Server Integration Services, the construction of a datacube to support multidimensional analyses through MDX queries and, finally, the creation of interactive dashboards to answer specific business needs.

1.1 Data understanding

In the **Tracks** dataset each record describes a single song and combines both textual features (title, lyrics, language and possible explicit language) and audio properties (bpm, loudness, spectral measures, rolloff, flux, rms, flatness, spectral complexity, pitch). It also contains discographic data, for example whether the track belongs to a single, an album or a compilation. Temporal information is available as well, such as the day, month and year in which the track was released. Finally, the dataset provides the number of streams one month after release, which is useful for analysing the commercial performance of the songs.
Each song in the Tracks dataset corresponds to a main artist and possibly one or more secondary artists. In particular, the main artist is linked to the **Artists** dataset through an identifier. This second dataset provides personal and geographical information about the artists, such as: name, gender, date and place of birth, nationality, years of activity and geographical coordinates. These data make it possible to link each song to its main author and to carry out comparative analyses across artists and geographical areas.

1.2 Data cleaning

During the data cleaning phase several missing values were identified and handled.
In the **Artist** dataset, the missing values of *birth_place*, *birth_date* and *active_start* were filled in manually based on external information, when available. The *country* attribute was standardised by assigning the value "Italia" to all artists, while the *nationality* column, redundant with respect to *country*, was removed. The missing values of *province* and *region* were derived from the place of birth through mapping dictionaries, while *latitude* and *longitude* were added using a geocoding service. Artists without geographical coordinates were assigned a null value. The *artist_type* variable was also introduced. In particular, for each artist the associated descriptive string was analysed: if the text contained the term "gruppo" (group), the artist was classified as *gruppo*; in all other cases, including the absence of a description, the artist was classified as *solista* (solo artist). Once this information had been extracted, the *description* column was removed from the dataset to avoid redundancy. The *active_end* variable was also removed, because it was not relevant for the planned analyses and had a high percentage of missing values, so it would not have contributed significantly to the analysis we were conducting.
In the **Tracks** dataset, the missing values of *year*, *month* and *day* were filled in using the album release date. Tracks that still had incomplete temporal information were excluded.
The missing values of *featured_artists* were extracted from the song title.
The *language* variable was first normalised where it was present but not standardised; where it was missing or unreliable, it was inferred automatically through language detection applied to the song lyrics.
The missing textual metrics (*n_tokens, n_sentences, char_per_tok, avg_token_per_clause*) were computed directly on the song lyrics. The missing values of the *explicit* attribute were derived from the presence of swear words. The missing audio values were imputed using the median of the corresponding audio feature computed at the level of each artist, exploiting the information available on the other tracks by the same artist.
In addition to identifying and handling null values, new attributes were created. The first one, called *geocode*, was obtained from the geographical coordinates in the Artists dataset. With this information it was possible to generate a geographical identifier to associate with each song.
The following identifiers were derived from the Artists and Tracks datasets. Some were created from scratch, while others were redefined to ensure consistency and uniqueness of the data:

- **geo_id**: assigned to each artist based on the place of birth. This identifier makes it possible to represent a single geographical area only once and to link several artists to the same territorial context.
- **id_track**: regenerated by combining the artist identifier, the album sequence and the position of the track within the album.
- **album_id**: regenerated by combining the artist identifier and a progressive index representing the sequence of albums within the artist's discography. In this way, all tracks belonging to the same album share the same *album_id*, while different albums by the same artist remain distinct.
- **text_id** and **symp_id**: created using the same hierarchical logic as *id_track*. Both identifiers combine the artist identifier, the album sequence and the position of the track.
- **date_id**: created in the integer format YYYYMMDD, obtained by combining the *year*, *month* and *day* fields.

1.3 Song profiling

In the Song Profiling process a new attribute was introduced in the final dataset, called *audio_label*, created to identify the category of each track based on the different audio features available. The tracks were grouped using the k-means clustering algorithm, and each cluster was then given a descriptive label, defined a posteriori based on the mean values of the audio features within each group. The labels assigned are: "Standard Urban", "Elettrico & Aggressivo", "Lo-Fi / Chill", "Mainstream Melodico" and "Banger".
We also investigated the possibility of obtaining music categories from the lyrics at our disposal. The initial goal was to identify thematic or stylistic categories based on the content of the lyrics, so as to differentiate the songs beyond purely structural information. In a first phase we applied a text clustering model to the song lyrics, evaluating the quality of the groupings through the silhouette score. The result showed a very low silhouette score, confirming the absence of distinct groups and suggesting little linguistic heterogeneity among the songs. We then tried an alternative approach, based on keyword classification: for each music category, lists of representative terms were defined and, through lexical search in the lyrics, songs were assigned to the corresponding category. This method also produced weak results: the lyrics have a very similar structure, and single songs were assigned to more than one class. Both approaches showed a common limitation: in this context, the lyrics do not provide enough variability to build distinctive profiles. For this reason we refocused the song profiling process exclusively on the intrinsic audio features of the songs.

1.4 Data Warehouse Schema

The Data Warehouse was designed to support the study of music streaming performance. The adopted schema (Figure 1) follows a star model, with the **FactSong** table as its central element. The granularity was defined at the level of the single music track ("id_track"), considering a snapshot one month after the release date. The attribute *streams_1month*, which represents the number of streams reached by a song in its first month after release, was selected as the measure. This metric was chosen because it is directly linked to the decision-support goals of the data warehouse and is fully additive with respect to the dimensions considered. Adopting a star schema guarantees structural simplicity and high performance in aggregation operations, supporting the execution of frequent analytical queries without introducing unnecessary join complexity.
The schema includes six dimension tables, each dedicated to describing a different aspect of the data. **DateDim** allows us to carry out temporal analyses, in particular thanks to the day, month, year and season attributes. It is identified by the attribute *date_id*, already built in the data cleaning phase from the complete temporal information of the track. **ArtistDim**, linked to **GeographyDim**, provides information about the artist and their geographical origin. The identifier *geo_id* was assigned during data preparation as a unique code associated with the artist's place of birth. The **GeographyDim** dimension therefore uses *geo_id* as its primary key to associate several artists with the same territorial context. **AlbumDim**, identified by the attribute *album_id*, contains descriptive information about the album. The identifier *album_id* was defined earlier. The tables

TextDim and **SymphonyDim**, identified by *text_id* and *symp_id* respectively, contain the textual and melodic information associated with the track. In this case too, the identifiers were assigned during data cleaning and then used to create the dimensions, ensuring a one-to-one correspondence with the tracks in the fact table.

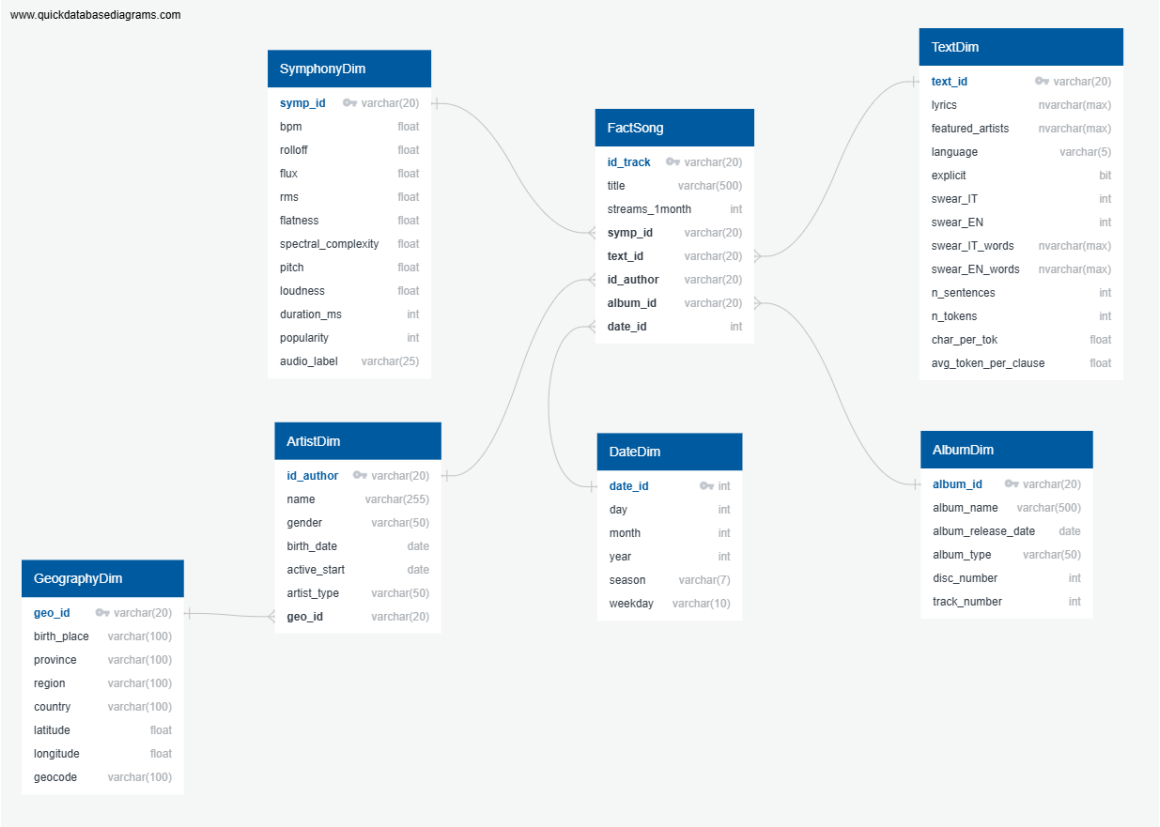


Figure 1: Data Warehouse schema

1.5 Splitting the data into the Data Warehouse tables

The source data were split into several CSV files, one for each Data Warehouse table defined in the design phase. The split was carried out by a Python program which, starting from the original datasets, extracts for each table only the attributes it needs, ensuring consistency with the adopted logical schema.
To create the dimension tables, a generic function was implemented that selects the unique records based on the primary key, avoiding data duplication and ensuring that the instances within each dimension are unique. This approach keeps a clear separation between dimensions and fact table, making the subsequent loading into the Data Warehouse easier.
The result of this phase is a set of structured and consistent input files, ready to be used both for programmatic loading with Python and in SSIS-based ETL processes.

1.6 Data uploading with Python

The data were loaded into the Data Warehouse in Python using the pyodbc library. A generic insertion function was defined and used for both the dimension tables and the fact table; it receives the table name, the input file and the columns to populate.
Before insertion, the data go through a technical cleaning step aimed at handling missing values and normalising numerical formats. During loading, consistency checks on the number of columns and uniqueness checks on the primary key are also performed, so as to prevent duplicate inserts or inserts that do not conform to the database schema.
Insertion is performed in batches of fixed size through the executemany operation, reducing the number of calls to the database and improving overall performance. The process is managed through explicit commits at the

end of each phase, allowing a rollback in case of error and guaranteeing the integrity of the loaded data.

1.7 Loading the data with SSIS (30% of the dataset)

The original Data Warehouse tables were replicated directly in SQL Server Management Studio by running an SQL query on each table.
This procedure made it possible to create tables structurally identical to the originals, but without records, containing only the schema.
The duplication was performed with the following query:

```
SELECT *
INTO table_name_SSIS
FROM table_name
WHERE 1 = 0;
```

To partially repopulate (30%) the duplicated tables, a solution in SQL Server Integration Services (SSIS) was implemented.
In an initial phase, the process completely deletes the records in the destination tables (SSIS tables), in order to guarantee that the load can be repeated and to avoid duplication or data inconsistency in subsequent runs.
Next, a percentage sampling of 30% was applied to the source data, allowing a random subset of records to be extracted. The sampled data are then routed to separate OLE DB Sources, each associated with a specific dimension.
To guarantee referential consistency with the fact table, a lookup step was implemented between the records of the sampled dimensions and the fact table sampled at 30%: only the records whose keys match are inserted into the corresponding SSIS tables, which are thus populated according to the required percentage.
An exception to this procedure is GeographyDim, for which the lookup is not performed on the fact table but on ArtistDim, since Geography is reached through the latter and not directly from the fact table.

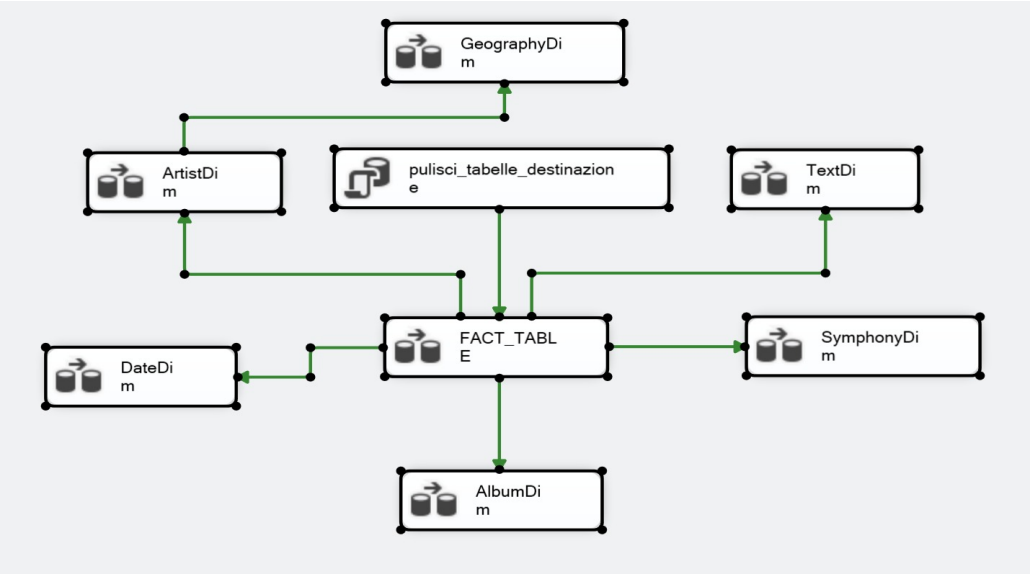


Figure 2: control flow assignment 7

2 Analytical queries with SSIS

2.1 Assignment 8

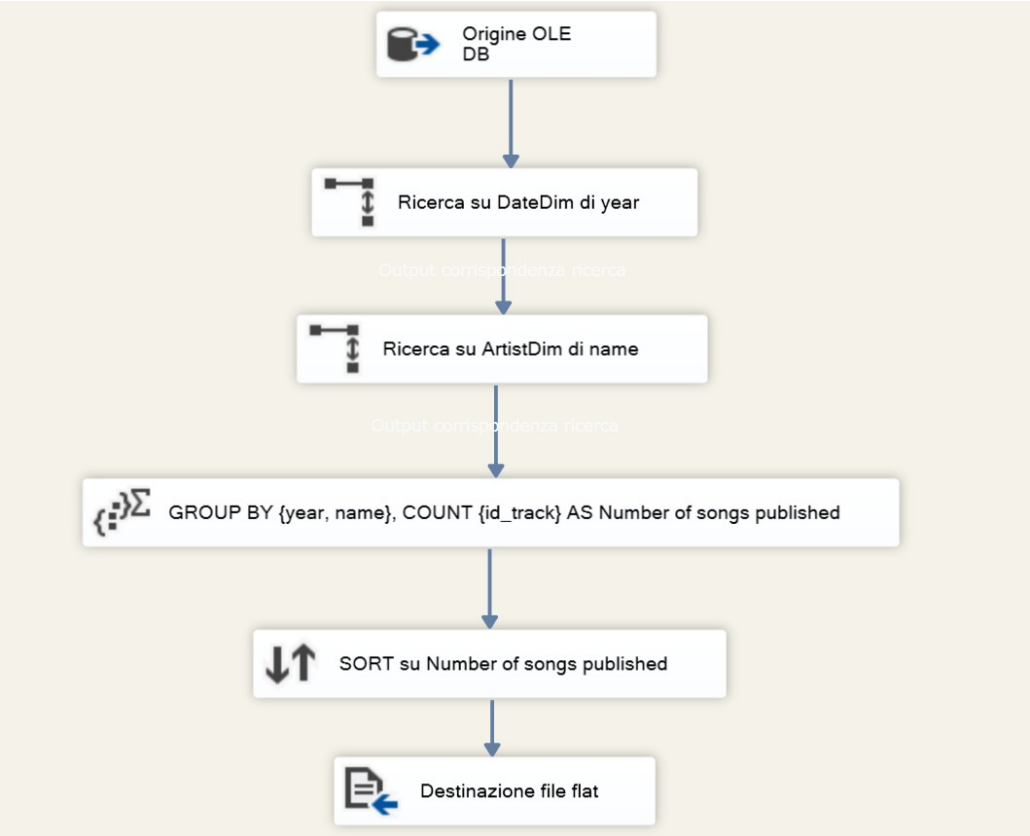


Figure 3: SSIS 8

To answer Assignment 8, a query was implemented in SSIS that lists, for each year, all the artists ordered by the total number of songs published.

2.2 Assignment 9

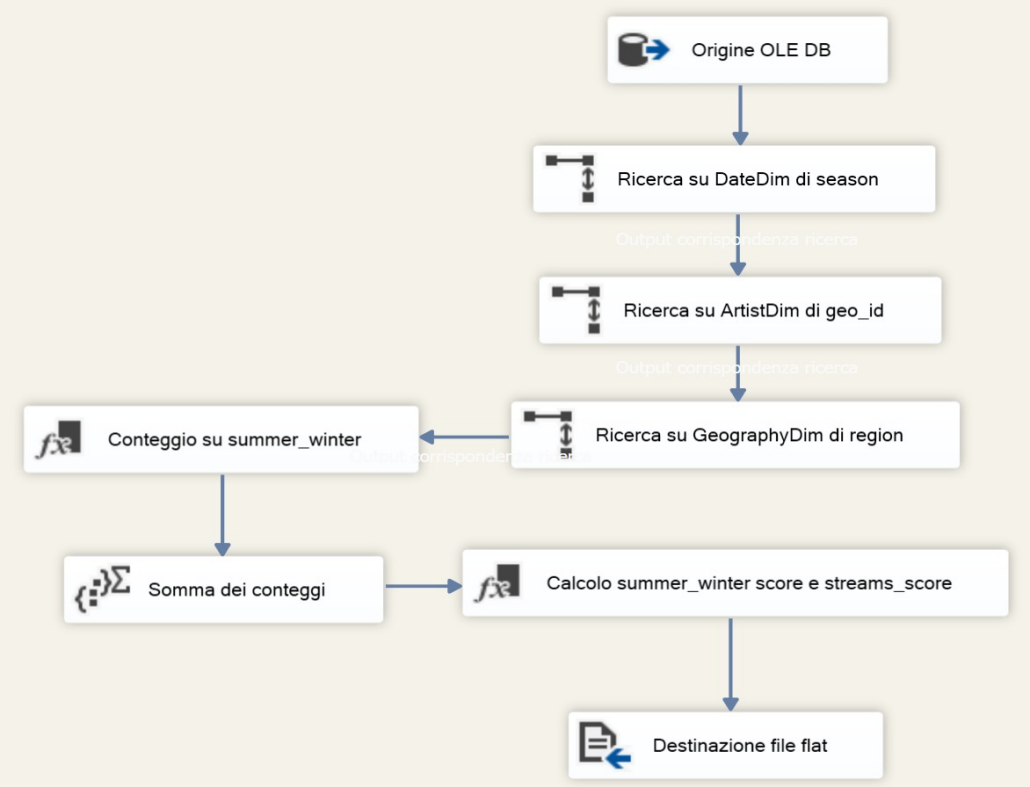


Figure 4: SSIS 9

For Assignment 9, in addition to the lookup operations on the temporal and geographical dimensions, some specific transformations were introduced to compute the required indicators. In particular, the box *Conteggio su summer_winter* (count on summer_winter) implements a conditional logic that classifies each song according to its release season, distinguishing between summer and winter based on the season attribute of DateDim. This classification makes it possible to count separately the number of songs released in the two seasons without using a multicast and several separate aggregations, reducing the duplication of the data flow and making the process more computationally efficient. Then the scores required by the assignment are computed: the summer–winter score (defined as the ratio between the number of songs released in summer and those released in winter) and the streams score (computed in the same way using the number of streams one month after release).

2.3 Assignment 10

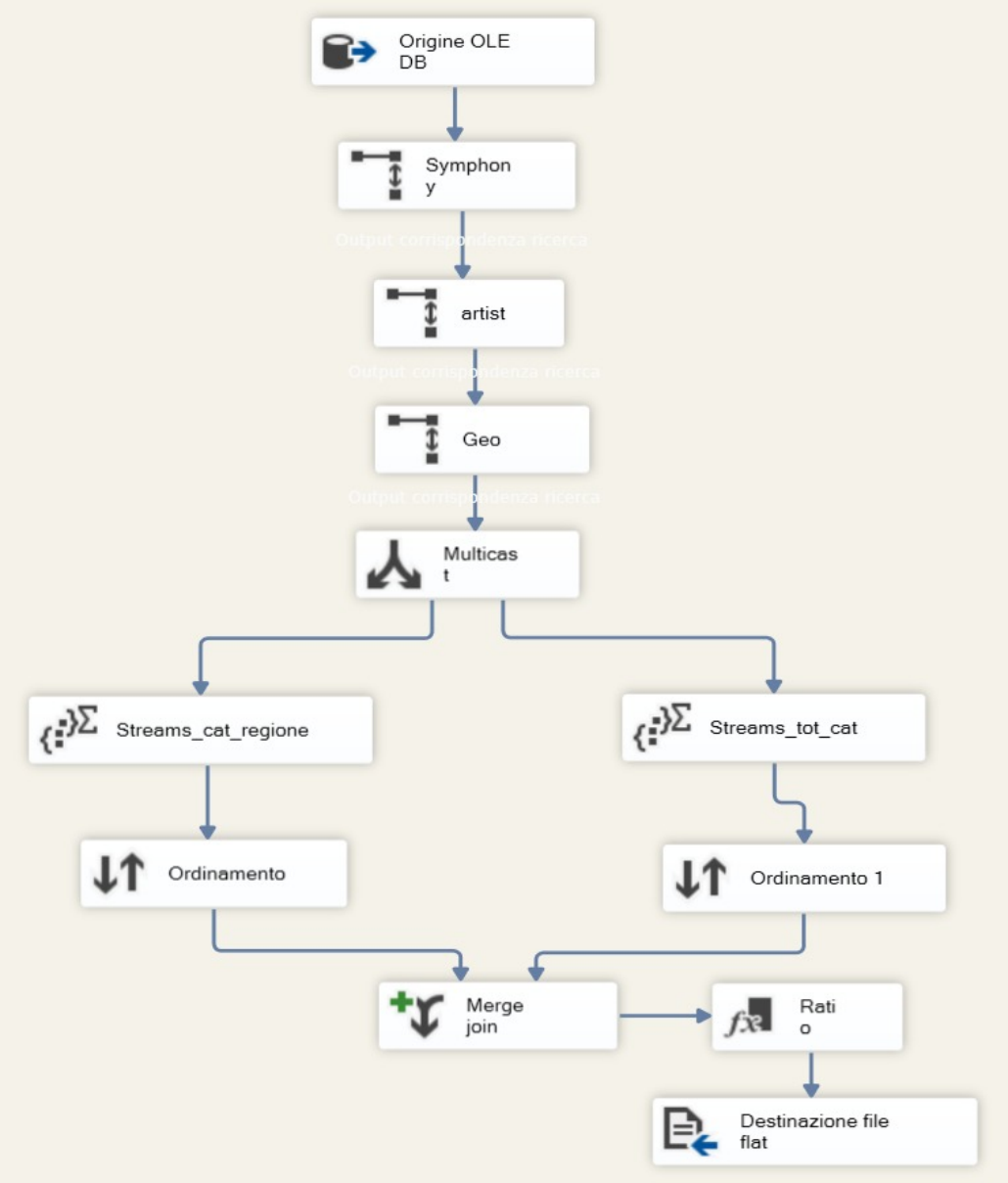


Figure 5: SSIS 10

To answer Assignment 10, an SSIS flow was built to compute indicators based on the distribution of streams by category and geographical area. During the data integration phase, records with a null value for the *region* attribute were excluded directly in the lookup transformation, so as to avoid propagating incomplete values into the subsequent aggregation operations. The data flow was then split through a Multicast into two separate branches: the first computes the streams aggregated by category and region, while the second computes the total streams for each category across all regions.
The two aggregated results are then combined through a Merge Join operation. In the final computation box the ratio is determined, defined as the ratio between the streams of the category in the region considered and the number of streams of the same category in the other regions, obtained by subtracting the regional streams from the category total.

2.4 Assignment 11

For Assignment 11 we chose to create a single *control flow* to compute both the *trending factor* and *trending percentage* metrics. This choice was motivated by the wish to define a single data extraction flow, avoiding redundant reads of the tables; moreover, this structure makes it possible to reuse intermediate results, such as the sums of trending songs, to feed both metrics at the same time.

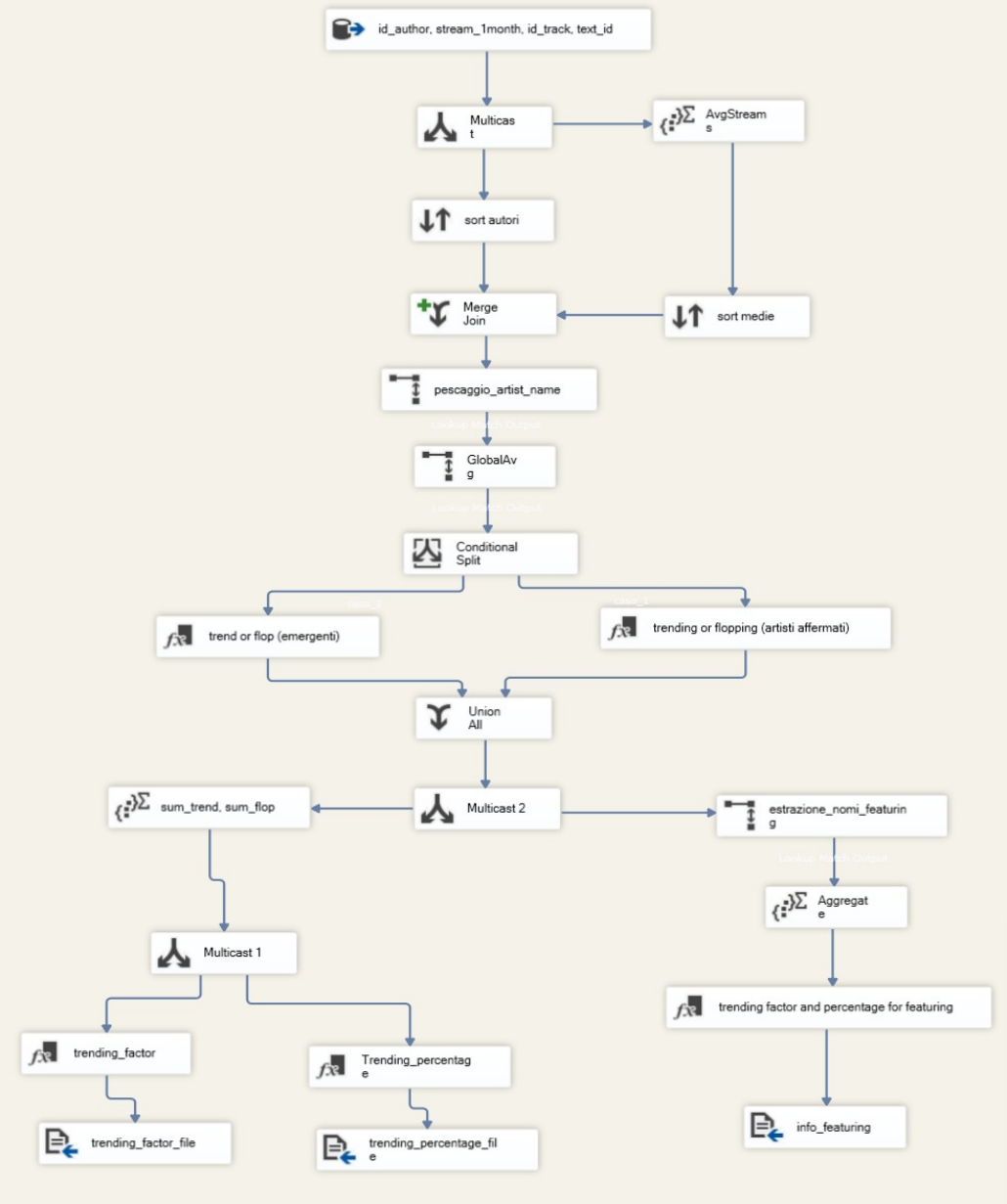


Figure 6: SSIS 11

As for the criterion chosen to decide whether a song should be considered trending or a flop, fixed percentage thresholds were used (25% above the average to classify a track as *trending* and 50% below the average to consider it *flopping*). This choice was made because, from a business point of view, it is more immediate and understandable for a stakeholder to say that a song is trending when it exceeds its own average by 25%. Moreover, this approach is robust to noise and outliers, since the threshold always remains tied to the artist's average performance: if the artist grows gradually, the average adapts accordingly, without being excessively influenced by isolated individual peaks.

To obtain more reliable results, two specific *edge cases* were considered, for which dedicated solutions were adopted:

- **false stable:** the category of artists who have released fewer than five songs, in particular those with only one song. In these cases, the average number of streams is statistically unreliable; therefore, the *trending factor* and the *trending percentage* were computed with reference to the global average of streams instead of the single artist's average.
- **famous hit:** the situation in which an artist has a song with a number of streams significantly higher than their usual performance, high enough to push the overall average upwards and classify as flops tracks that, in absolute terms, should not be. To handle this case an additional threshold was introduced in the classification process: a song that exceeds 8000 streams in one month is automatically considered successful, even if it does not exceed the artist's average by 25%.

2.5 Assignment 12

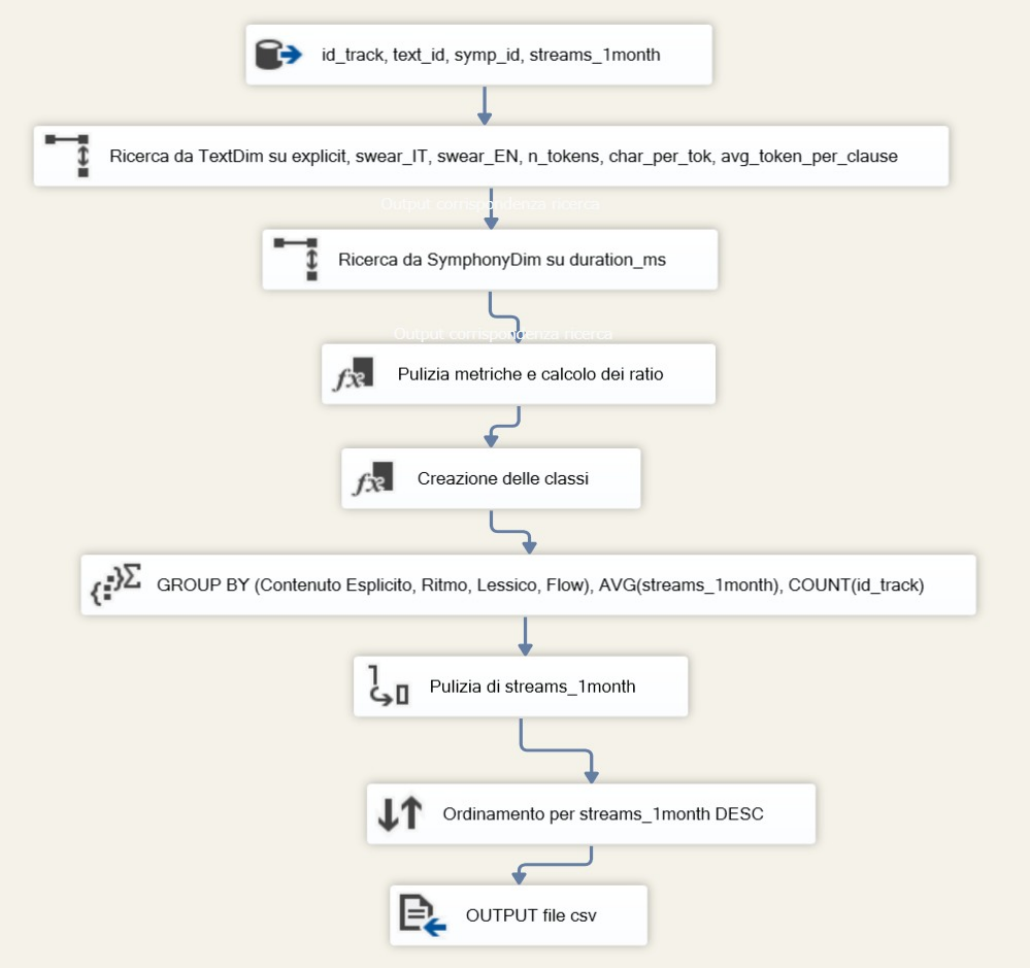


Figure 7: SSIS 12

For Assignment 12 an analytical query was defined to study the relationship between some textual and musical features of the songs and their performance in terms of streams. In particular, attributes from the TextDim and SymphonyDim dimensions were used, then combined with the *streams_1month* measure.
In a first step, the ratio between the number of swear words and the total number of words in the lyrics was computed and used to classify the songs into three categories of explicit content: clean, with a few swear words and with many swear words. Similarly, three classes based on the length of the lyrics were defined: short, medium and long. Finally, three classes were obtained from the attribute *char_per_tok*, used as an indicator of the lexical complexity of the song: basic lexicon, standard lexicon and rich lexicon.
These dimensions are complemented by a rhythm classification, derived from the musical attributes.
Once the classes had been defined, the songs were aggregated through a group by on the categories of explicit

content, rhythm, lexicon and lyrics structure. For each combination, the average number of streams one month after release and the number of tracks were computed, making it possible to analyse which combinations of features are associated with better performance.

2.6 Assignment 13

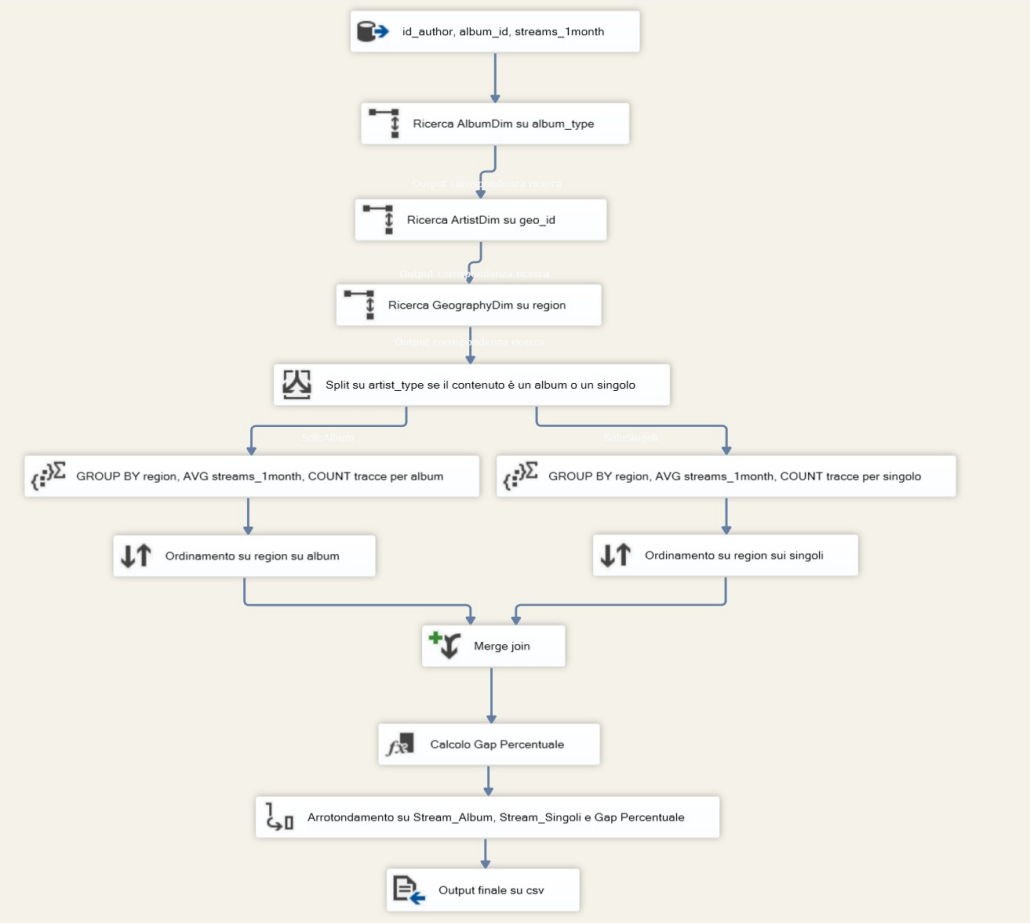


Figure 8: SSIS 11

The proposed query aims to compare the streaming performance of albums and singles across the different regions, analysing any significant differences in terms of average number of streams. This type of analysis is relevant for a record label because it makes it possible to assess whether, and in which geographical areas, releasing singles is more effective than distributing full albums, supporting strategic marketing decisions and release planning.

The flow starts by extracting information on artists, albums and the number of streams one month after release, which is then enriched through lookup operations on the AlbumDim, ArtistDim and GeographyDim dimensions. The data are then separated into two distinct branches through a split based on the type of content, distinguishing between songs belonging to albums and singles. For each branch, the average number of streams and the number of tracks per region are computed through aggregation. The results are then sorted and combined through a Merge Join operation. In the final step of the flow the percentage gap between the average performance of albums and singles is computed, in order to highlight any significant differences at regional level.

3 Multidimensional Data Analysis

3.1 OLAP Cube

To create the multidimensional data cube, an **Analysis Services Multidimensional and Data Mining** solution was created in Visual Studio, in which the following operations were performed:

1. Connection to our DW Group_ID_6 as data source
2. Creation of a view to define the relationships between the tables
3. Creation of the dimensions; in particular, a selection of the attributes of each table was made (also with a view to the required **MDX** queries), with the exception of AlbumDim, which was discarded entirely because it was judged not useful for the purpose
4. Definition of the hierarchies for the temporal dimension (DateDim) and for the geographical dimension (GeographyDim)
5. Creation and subsequent deployment of the multidimensional cube on SSMS in order to run MDX queries

The only measure included in the cube is the number of streams one month after the release of the song.

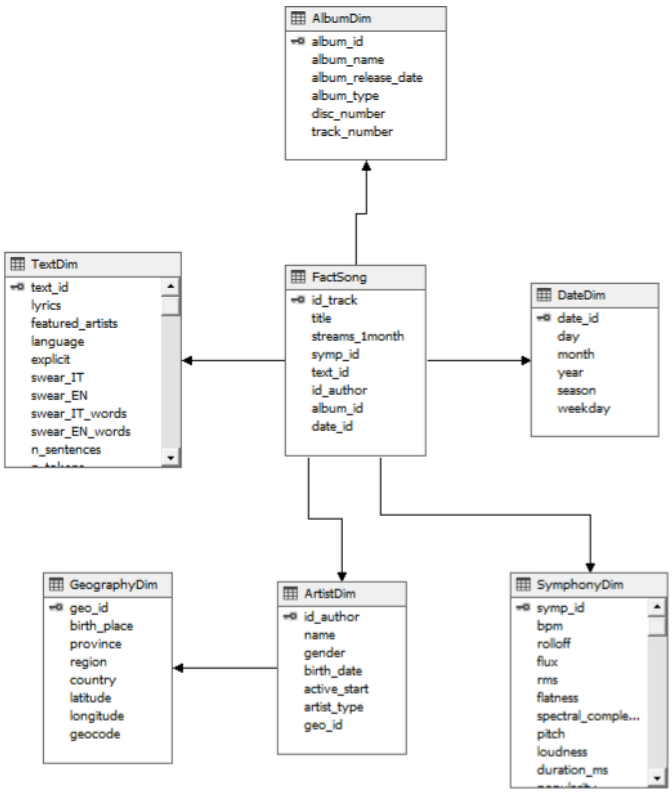


Figure 9: view of the data cube in Visual Studio

3.2 MDX queries

3.2.1 Assignment 15

```
SELECT
    NON EMPTY {
        EXCEPT(
            [Date Dim].[Year].[Year].Members,
            { [Date Dim].[Year].[Unknown] }
        )
```

```
        *
        {
            [Date Dim].[Month].[Month].Members,
            [Date Dim].[Month].[All]
        }
    } ON COLUMNS,

    NON EMPTY {

        FILTER(
            [Geography Dim].[Region].Children,
            Len([Geography Dim].[Region].CurrentMember.Name) > 0
        ),

        { [Geography Dim].[Region].[All] }

    } ON ROWS

FROM [Group ID 6 DB]
WHERE ( [Measures].[Streams 1month] )
```

3.2.2 Assignment 16

```
WITH
MEMBER [Measures].[Score Main] AS
    [Measures].[Streams 1month] * 0.8

MEMBER [Measures].[Score Featured] AS
    SUM(
        FILTER(
            NONEMPTY(
                [Text Dim].[Featured Artists].Members,
                ( [Measures].[Streams 1month], [Artist Dim].[Name].[All] )
            ),
            InStr(
                1,
                [Text Dim].[Featured Artists].CurrentMember.Name,
                CASE
                    WHEN InStr(1, [Artist Dim].[Name].CurrentMember.Name, "guè pequeno", 1) > 0 THEN "Guè"
                    WHEN InStr(1, [Artist Dim].[Name].CurrentMember.Name, "anna pepe", 1) > 0 THEN "Anna"
                    WHEN InStr(1, [Artist Dim].[Name].CurrentMember.Name, "chadia rodriguez", 1) > 0 THEN "Chadia"
                    WHEN InStr(1, [Artist Dim].[Name].CurrentMember.Name, "samuel heron", 1) > 0 THEN "Samuel"
                    ELSE [Artist Dim].[Name].CurrentMember.Name
                END,
                1
            ) > 0
        ),
        ( [Measures].[Streams 1month], [Artist Dim].[Name].[All] )
    ) * 0.2

MEMBER [Measures].[Total Year Streams] AS
    (
        [Measures].[Streams 1month],
        [Artist Dim].[Name].[All],
        [Text Dim].[Featured Artists].[All]
    )

MEMBER [Measures].[Avg Yearly Artist Streams] AS
    DIVIDE(
        ( [Measures].[Score Main] + [Measures].[Score Featured] ),
        [Measures].[Total Year Streams]
```

```
    ),
    FORMAT_STRING = 'Percent'

SELECT
    NON EMPTY { [Date Dim].[Year].[Year].Members } ON COLUMNS,
    NON EMPTY { [Artist Dim].[Name].[Name].Members } ON ROWS
FROM [Group ID 6 DB]
WHERE ( [Measures].[Avg Yearly Artist Streams] )
```

To ensure a correct attribution of listening volumes to the artists who appear in collaborations, a validation procedure was implemented to identify the artist aliases that did not match. These were then handled in MDX, in order to obtain a weighted streams metric and provide an accurate view of that artist's influence on the market.

3.2.3 Assignment 17

```
WITH
    -- 1. Retrieve the streams of the previous year
    MEMBER [Measures].[Streams Prev Year] AS
    (
        [Measures].[Streams 1month],
        [Date Dim].[Year].CurrentMember.PrevMember
    )

    -- 2. Compute the percentage change (Growth)
    MEMBER [Measures].[YoY Growth] AS
    DIVIDE(
        ([Measures].[Streams 1month] - [Measures].[Streams Prev Year]),
        [Measures].[Streams Prev Year]
    ), FORMAT_STRING = "Percent"

SELECT
    -- COLUMNS: Years
    NON EMPTY {
        EXCEPT(
            [Date Dim].[Year].[Year].Members,
            { [Date Dim].[Year].[Unknown] }
        )
    } ON COLUMNS,

    -- ROWS: The categories
    NON EMPTY {
        [Symphony Dim].[Audio Label].[Audio Label].Members
    } ON ROWS

FROM [Group ID 6 DB]
WHERE ( [Measures].[YoY Growth] )
```

The analysis highlighted anomalous percentage growth rates, concentrated mostly in the first years of observation. This distortion means that a negligible number of streams in the previous year generates exponential percentages, amplified if in the following year a song that was successful in that profile is recorded. A contextualisation is therefore needed to distinguish the actual explosion of a trend from mere statistical noise.

3.2.4 Assignment 18

```
WITH
MEMBER [Measures].[Avg Prev Season] AS
```

```
    AVG(
        DESCENDANTS(
            ParallelPeriod(
                [Date Dim].[Year].[Year],
                1,
                [Date Dim].[Year].CurrentMember
            ),
            [Date Dim].[Month]
        ),
        [Measures].[Streams 1month]
    )

MEMBER [Measures].[Season Increase Percentage] AS
    IIF(
        ISEMPTY([Measures].[Avg Prev Season])
        OR [Measures].[Avg Prev Season] = 0,
        NULL,
        (
            [Measures].[Streams 1month]
            - [Measures].[Avg Prev Season]
        )
        / [Measures].[Avg Prev Season]
    ),
    FORMAT_STRING = "0.00%"

SELECT
{
    [Measures].[Streams 1month],
    [Measures].[Avg Prev Season],
    [Measures].[Season Increase Percentage]
} ON COLUMNS,

NON EMPTY
FILTER(
    [Symphony Dim].[Audio Label].[Audio Label].Members
    *
    [Date Dim].[Season].[Season].Members,
    [Measures].[Streams 1month] > [Measures].[Avg Prev Season]
) ON ROWS

FROM [Group ID 6 DB]
```

The computation compares the monthly streams of the current season with the average of the same season in the previous year, highlighting only the cases of positive growth. However, when historical volumes are low, even small absolute changes can translate into high percentage increases, so the result should be read with caution.

3.2.5 Relative popularity index

This analysis, implemented through an MDX query, aims to determine the degree of specialisation of regional music consumption with respect to the national benchmark. The computed index compares the local market share of a given genre with its average incidence on total national streams.

```
MEMBER [Measures].[Calc_Region_Share] AS
    DIVIDE(
        [Measures].[Streams 1month],
```

```
        ([Measures].[Streams 1month], [Symphony Dim].[Audio Label].[All]),
        0
    ), FORMAT_STRING = "Percent"

MEMBER [Measures].[Calc_Global_Share] AS
    DIVIDE(
        ([Measures].[Streams 1month], [Geography Dim].[Hierarchy].[All]),
        ([Measures].[Streams 1month], [Geography Dim].[Hierarchy].[All], [Symphony Dim].[Audio Label].[All]),
        0
    ), FORMAT_STRING = "Percent"

MEMBER [Measures].[Calc_Penetration_Index] AS
    IIF(
        [Measures].[Calc_Global_Share] = 0,
        0,
        [Measures].[Calc_Region_Share] / [Measures].[Calc_Global_Share]
    ), FORMAT_STRING = "0.00"

SELECT
{
    [Measures].[Calc_Region_Share],
    [Measures].[Calc_Global_Share],
    [Measures].[Calc_Penetration_Index]
} ON COLUMNS,
NON EMPTY
(
    [Geography Dim].[Hierarchy].[Region].Members
    * [Symphony Dim].[Audio Label].[Audio Label].Members
) ON ROWS
FROM [Group ID 6 DB]
```

A Relative Popularity Index value above 1.00 indicates a positive anomaly: in that specific area, the genre is more relevant than the national average. The results show empirical evidence of strong territorial cultural clusters:

- The genre **"Mainstream Melodico"** in **Calabria** has an index of **2.80**, indicating a penetration almost three times the average.
- The genre **"Lo-fi/Chill"** in **Toscana** reaches a value of **2.09**, highlighting a regional preference for atmospheric sounds.

Business Insight: From a strategic point of view, the RPI acts as an indicator of *Market Attractiveness*. These data allow record labels and promoters to optimise the allocation of the advertising budget and to plan live tours with reduced business risk, concentrating resources where organic demand is statistically highest and loyalty to the genre is already established.

4 Dashboards

4.1 Distribution of streams by artist's place of birth and song category

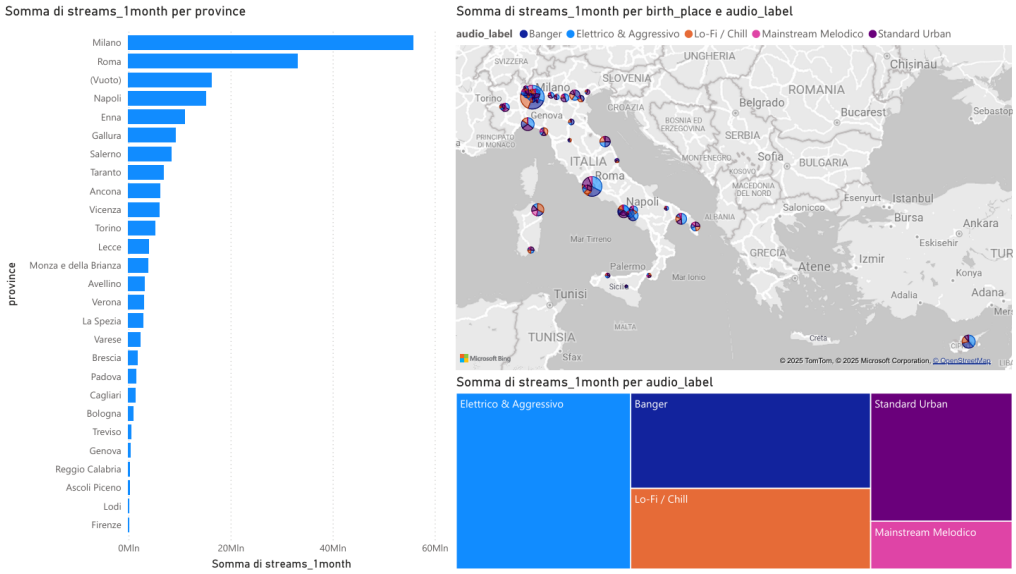


Figure 10: Assignment 20

This dashboard was designed to analyse the temporal evolution of music consumption, relating the volume of streams to the audio categories and the type of artist.

The **Line Chart** is the main choice for visualising the trend of streams over time, since it clearly captures the growth and decline dynamics of the different audio labels over the years. Using multiple lines allows a direct comparison between the music categories, showing how some of them, in particular Banger and Elettrico & Aggressivo, experienced much stronger growth from the mid-2010s onwards. This visualisation is particularly effective at highlighting both popularity peaks and structural changes in the music market.

To enrich the analysis, a **Stacked Bar Chart** was added showing the count of audio labels per year. This choice makes it possible to observe the evolution of music production over time, giving an indication of the number of tracks released in each period and allowing the stream trends to be interpreted also in terms of the amount of content available.

The **Donut Chart** dedicated to the type of artist (solo vs group) was introduced to offer an immediate summary of the distribution of streams between the two categories. The predominance of solo artists emerges clearly, suggesting a greater concentration of commercial success on individual artists rather than on bands.

Finally, the **Treemap** makes it possible to compare the overall weight of the different audio labels in terms of total streams. This visualisation is particularly suited to representing proportions and hierarchies, intuitively highlighting the dominant music categories within the dataset.

Overall, the combination of these visualisations makes it possible to analyse the phenomenon of music streaming from several perspectives (temporal, categorical and structural), offering a complete and coherent reading of the dynamics that characterise the evolution of the music market on streaming platforms.

4.2 Lyrics analysis

The analysis of textual complexity through a **Line Chart** reveals that the average length of the lyrics does not follow a linear growth trend, and instead shows strong volatility over the years. A particularly relevant insight emerges when drilling down on the monthly time attribute: while the output of solo artists keeps an extremely constant and predictable lyrics length, bands show a much higher variance. This discrepancy suggests that the creative dynamics of bands are less standardised, probably influenced by the number of contributors or by the episodic nature of collective projects compared with the regularity of individual artists.
For the second visualisation a **Stacked Area Chart** was chosen to analyse the volume of streams according to the nature of the lyrics (Explicit vs Clean) over the years. The chart shows not only that songs with explicit language generate a higher volume of sales, but also reveals a clear structural change in the market: while until 2010 the share of streams between explicit and non-explicit content was balanced, in the last decade there has been an almost total dominance of 'Explicit' songs. This trend, which keeps growing, suggests that explicit language has become an intrinsic feature of the dominant genres on streaming platforms, directly influencing commercial performance.
The last two visualisations are made with a **Stacked Bar Chart**. The first analyses the artists' propensity to use explicit language, distinguishing between swear words in Italian and in English. A strong tendency towards the internationalisation of language clearly emerges: for several artists, the use of foreign explicit terms exceeds that of local ones, reflecting the overseas influences of the genre. In parallel, the second chart segments the explicit tracks by music genre, showing that the songs classified as 'Banger' and 'Lo-Fi / Chill' are the most explicit in the dataset. This cross-analysis suggests that explicitness is not limited to aggressive genres, but has become a pervasive component even of more relaxed sounds, consolidating a communication model common to the whole modern music scene.

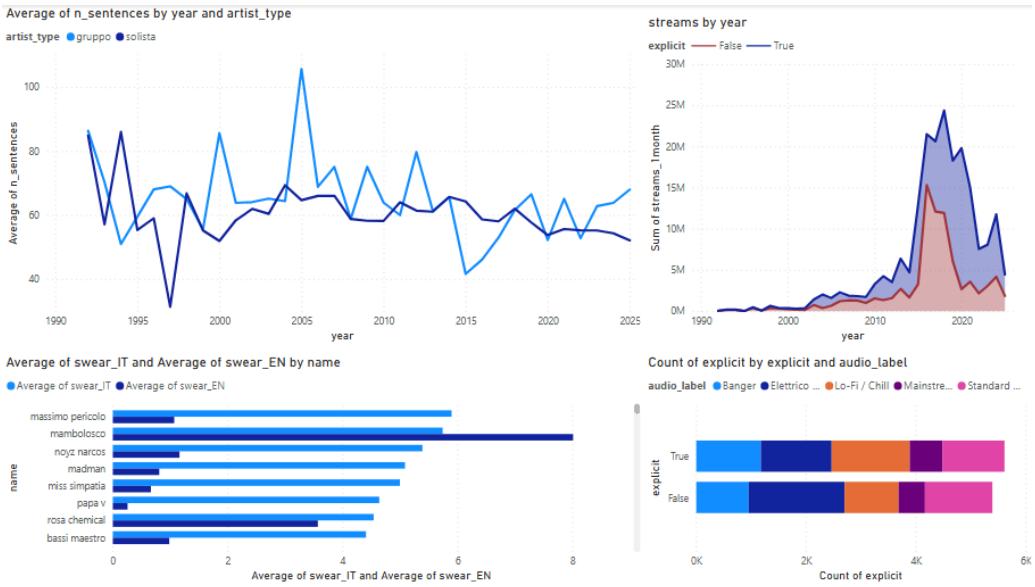


Figure 11: Assignment 21

4.3 Temporal evolution of streams and music categories

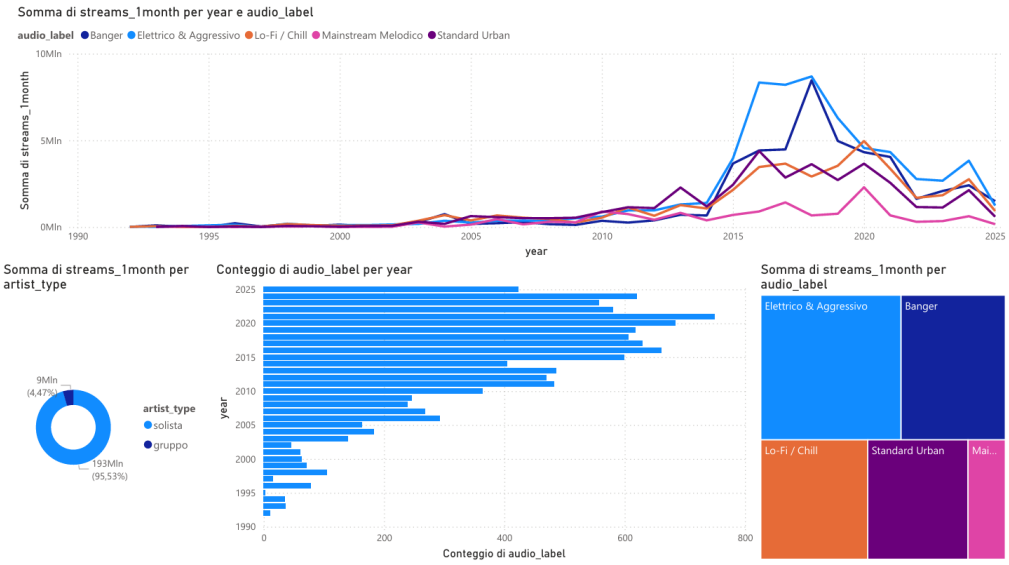


Figure 12: Assignment 22

This dashboard was designed to analyse the temporal evolution of music consumption, relating the volume of streams to the audio categories and the type of artist.

The **Line Chart** is the main choice for visualising the trend of streams over time, since it clearly captures the growth and decline dynamics of the different audio labels over the years. Using multiple lines allows a direct comparison between the music categories, showing how some of them, in particular Banger and Elettrico & Aggressivo, experienced much stronger growth from the mid-2010s onwards. This visualisation is particularly effective at highlighting both popularity peaks and structural changes in the music market.

To enrich the analysis, a **Stacked Bar Chart** was added showing the count of audio labels per year. This choice makes it possible to observe the evolution of music production over time, giving an indication of the number of tracks released in each period and allowing the stream trends to be interpreted also in terms of the amount of content available.

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Overall, the combination of these visualisations makes it possible to analyse the phenomenon of music streaming from several perspectives (temporal, categorical and structural), offering a complete and coherent reading of the dynamics that characterise the evolution of the music market on streaming platforms.